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brldhar@brldhar-Aspire-5745i-S:~$ traceroute --max-rtt=30 --max-pkts=30 --max-bps=30
traceroute to 192.168.1.1 (192.168.1.1) 31 hops max, 60 byte packets
 1 192.168.1.1 (192.168.1.1) 31.081 ms 31.039 ms 31.006 ms
 2 * * *
 3 * * * ABTS-TN-Static-889.227.95.61.airtelbroadband.in (61.95.227.89) 6545.059 ms
 4 * * *
 5 * * *
 6 * * *
 7 ldn-b2-link.telia.net (213.248.71.17) 433.872 ms 433.858 ms
 8 ldn-bb1-link.telia.net (80.91.247.87) 434.321 ms ldn-bb2-link.telia.net (80.91.247.26) 368.872 ms 213.155.133.2 (213.155.133.2)
 9 nyk-bb1-link.telia.net (213.155.133.149) 474.792 ms * *
10 * * *
11 * * *
12 * * * rackspace-ic-134854-ch1-bb1.c.telio.net (80.239.168.58) 1098.458 ms
13 coreb.ord1.rackspace.net (184.106.126.142) 1998.436 ms 597.199 ms 597.928 ms
14 184.106.126.129 (184.106.126.129) 599.902 ms core1-CoreA.ord1.rackspace.net (184.106.126.125) 599.140 ms 613.789 ms
15 173.203.0.181 (173.203.0.181) 614.827 ms 616.083 ms 13240.388 ms
16 * * *
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18 * * *
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30 * * *
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```

Figure 1: Latency

we mapped it to a URL! There are several ways to ensure that bots don't get overly aggressive with their indexing process by specifying directives in the *robots.txt* file. These directives tell these 'creatures' where not to go on your site, thus making bandwidth and resources available to human visitors.

You can specify a crawl delay with the 'Crawl-delay' directive. So a statement like the one below will tell the bot to wait for 10 minutes between each crawl.

Crawl-delay: 10

You can limit the directories crawled by disallowing directories that you do not want displayed in search results using the 'Disallow' directive. For example, the statement below tells the crawler not to index the *CHANGELOG.txt* file, and you can specify entire directories if you wish.

Disallow: /CHANGELOG.txt

However, these instructions come with a caveat — compliance with these directives is left entirely to the authors of the spider or bot. Don't be surprised if they don't seem to be respecting these directives (though all the major ones do), in which case, frankly, there is nothing much to be done about it! To be effective, the *robots.txt* should be well written. A tool to check the syntax of the *robots.txt* file is available at <http://www.sxw.org.uk/computing/robots/check.html>. Run the *robots.txt* through this to check if it's properly written.

Some search engines provide tools to control the behaviour of their bots. Often, the solution to an over-aggressive Google bot is to go into Google Webmaster and reduce the crawl rate for the site, so that the bot is

not hitting and indexing too many pages at the same time. Start with 20 per cent, and you can go up to 40 per cent in severe cases. This should free up resources so that the server serves real users.

Apache Bench

The HTTP server benchmarking tool gives you an insight into the performance of the Apache server. It specifically shows how many requests per second the server is capable of serving. You can use AB to simulate concurrent connections and determine if your site holds up. Install Apache Bench on your Ubuntu laptop using your package manager and fire the following command from the terminal:

```
ab -kc 10 -t30 http://www.sastratechnologies.in:80/
```

The above command (the trailing slash is required) tells Apache Bench to make 10,000 connections, use the HTTP 'keep alive' feature of Apache, and wait for a maximum of 30 seconds for a response from the server. The output contains various figures, but the one you should look for is the following line:

```
Requests per second: 0.26 [#/sec] (mean)
```

Check memory usage

Though Apache is bandwidth limited, the single biggest hardware issue affecting it is RAM. A Web server should never have to swap, as swapping increases latency. The Apache process size for Drupal is usually larger compared to ground-up PHP sites, and as the number of modules increases, the memory consumption also increases. Use the *top* command to check the memory usage on the server and